Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Child Health

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Title: Management of Infants with Critical Pertussis

Purpose of this document :

Review the indications for admission to high dependency, Pediatric ICU units and review the management of severe cases of pertussis

Introduction :

Pertussis (whooping cough) is caused by Bordetella pertussis and is associated with severe disease in unvaccinated infants. Typical pertussis is characterized by severe paroxysmal coughing that can persist for weeks after initial onset. Severe Pertussis in young infants can manifest with apnea , encephalopathy, pneumonia, marked leukocytosis and pulmonary hypertension.

Critical pertussis is defined as pertussis disease that results in PICU admission or death and fatal Pertussis is associated with extreme leukocytosis and pulmonary hypertension.

Risk factors for critical disease include:

- Age group less than 3 months
- Presence of pneumonia
- High white cell count > 50 cell/mm^3
- Evidence of pulmonary hypertension

Diagnosis

- Culture or PCR on nasopharyngeal specimen (swab or aspirate). Nasopharyngeal PCR is considered gold standard and is used for diagnosis in our hospitals.
- Leukocytosis with lymphocytosis (white blood cell count of ≥ 20 cells/mm3 with ≥ 50% lymphocytes) in infant with coryza, cough or apnea strongly indicate pertussis.

Indications for admission to medical high dependency unit:

- Monitoring for apnea in absence of pneumonia or hyper-leukocytosis

Indications for admission to PICU

- Pneumonia with need for respiratory support either in the form of high flow nasal cannula(HFNC), non-invasive ventilation(NIV) or invasive ventilation
- Frequent significant apnea
- Presence of shock and multi organ failure including pulmonary hypertension
- Rapid increase in WBC count with clinical deterioration

Antibiotic

Suspected cases of pertussis should be started on macrolide treatment without waiting for laboratory confirmation(see table 1).

Treatment of choice in small infants is Azirthromycin with daily dose of 10 mg/kg for 5 days. Azithromycin rather than erythromycin is recommended for young infants because of association of hypertrophic pyloric stenosis with erythromycin use.

TABLE 1 Recommended antimicrobial treatment and postexposure prophylaxis for pertussis, by age group

	Primary agents			
Age group	Azithromycin	Erythromycin	Clarithromycin	Alternate agent, TMP-SMZ[a]
<1 mo	Recommended agent. 10 mg/kg per day in a single dose for 5 days (limited safety data available)	Not preferred. Associated with infantile hypertrophic pyloric stenosis. Use if azithromycin is unavailable; 40–50 mg/kg per day in 4 divided doses for 14 days	Not recommended (safety data unavailable)	Contraindications for infants ages <2 mo (risk for kernicterus)
1–5 mo	10 mg/kg per day in a single dose for 5 days	40–50 mg/kg per day in 4 divided doses for 14 days	15 mg/kg per day in 2 divided doses for 7 days	Contraindicated at age <2 mo. For infants aged >2 mo, TMP at 8 mg/kg per day and SMZ at 40 mg/kg per day in 2 divided doses for 14 days
Infants (aged >6 mo) and children	10 mg/kg in a single dose on day 1 and then 5 mg/kg per day (maximum: 500 mg) on days 2–5	40–50 mg/kg per day (maximum: 2 g per day) in 4 divided doses for 14 days	15 mg/kg per day in 2 divided doses (maximum: 1 g per day) for 7 days	TMP at 8 mg/kg per day and SMZ at 40 mg/kg per day in 2 divided doses for 14 days
Adults	500 mg in a single dose on day 1 then 250 mg per day on days 2–5	2 g per day in 4 divided doses for 14 days	1 g per day in 2 divided doses for 7 days	TMP at 320 mg per day and SMZ at 1,600 mg per day in 2 divided doses for 14 days

[a]Trimethoprim-sulfamethoxazole (TMP-SMZ) can be used as an alternate to macrolides in patients aged ≥2 months who are allergic to macrolides, who cannot tolerate macrolides, or who are infected with a rare macrolide-resistant strain of *Bordetella pertussis*.

Reproduced from Nieves DJ, Heininger U. Bordetella pertussis. Microbiol Spectr. 2016 Jun; 4(3). doi: 10.1128/microbiolspec.EI10-0008-2015. PMID: 27337481

Intensive Care Management In Critical Pertussis

- Apnea should be closely monitored for need of invasive ventilation. Isolated apnea in the absence of pneumonia and pulmonary hypertension has excellent prognosis.
- Proper hydration and nutrition with target fluid intake at maintenance requirement
- Chest physiotherapy and suction for clearance of secretions
- Symptomatic treatment for seizures
- Use HFNC and NIV as first line modalities for mild-moderate cases of respiratory distress. Continuous monitoring for treatment success is needed. Features of non-response to HFNC and NIV are persistence of tachypnea, tachycardia and increase in the work of breathing. Invasive ventilation is indicated in respiratory failure, shock and failure of HFNC and NIV .

Invasive mechanical ventilation should include use of lung protective strategies with use of low-tidal volume (6-8 mL/kg) strategy and permissive hypercarbia. Ventilated Patients requiring mean airway pressure of 16 cm H2O and peak pressure of 30 cm H2O and more to achieve adequate oxygenation are indicated for use of high-frequency oscillatory ventilation.

- Exchange transfusion can reduce mortality if performed early before onset of organ failure(see indications for exchange transfusion in the text below)
- Cardiac assessment with ECHO in patients needing hemodynamic support and exchange transfusion

- Use of nitric oxide has limited value in pulmonary hypertension related to critical pertussis

Exchange Transfusion

Mortalities in pertussis was found to be related to hyper-leukocytosis and pneumonia. Use of exchange transfusion in pertussis- related leukocytosis is associated with reduced risk of mortality provided it is done early in the disease and before onset of organ failure.

<u>Indications for Exchange Transfusion</u>:

- WBC of ≥ 30 /mm^3 and increasing in presence of pneumonia, respiratory distress and tachycardia (> or = 180/min)
- WBC of ≥ 30 /mm^3 and presence of pulmonary hypertension, hemodynamic instability
- WBC > 100/mm^3 regardless of presence or absence of other clinical manifestations of critical pertussis

WBC should be checked every 6 hours if increasing and once per day if stable and not increasing.

<u>How to perform exchange transfusion</u>:

Amount of volume for exchange: double the patient's circulating blood volume

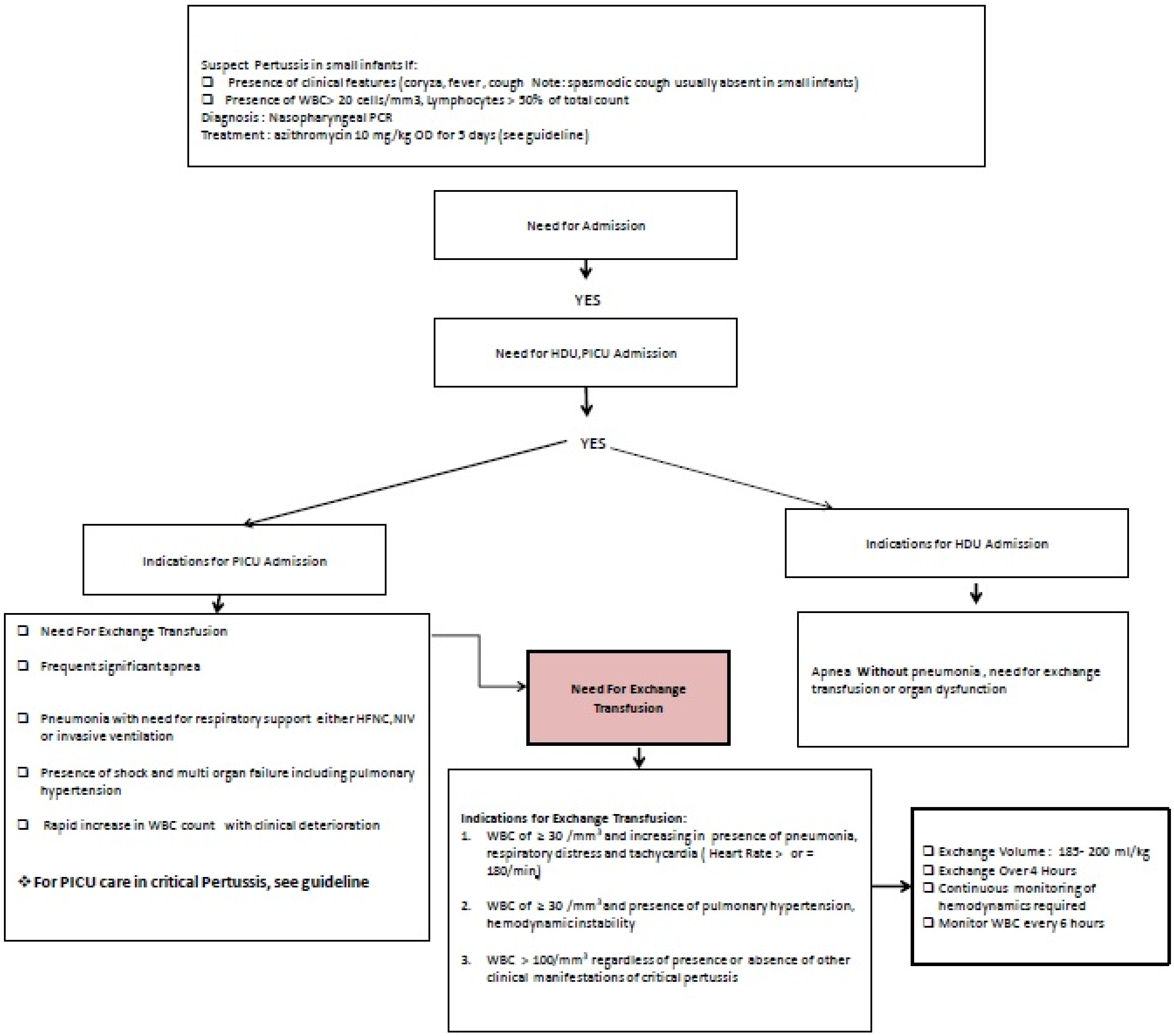
Total volume for exchange = 2x 85 ml/kg (~200ml/kg) performed over 4 hours

The amount of blood can be reconstituted to target final hematocrit 40-45. Combination of 4.5% albumin and packed red cells can be used to target hematocrit 40-45.

Vascular access for exchange transfusion: arterial and good venous catheters including central line to withdraw and infuse blood.

Exchange should be in aliquots of 10-20 ml depending on infant's hemodynamic stability. Continuous monitoring of vital signs is essential.

Monitoring : for secondary hypo-calcemia and hypomagnesaemia is required.



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